noks -> N = 18
a = 1
b = 8

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② Enkripsi Cara Logo Toko dengan Matriks Invers 3x3

a) Matriks A

$$A = \begin{bmatrix} 1+1 & 8 & 2 \\ 1 & 1+2 & 8 \\ 8 & 1 & 1+3 \end{bmatrix}$$

$$A = \begin{bmatrix} 2 & 8 & 2 \\ 1 & 3 & 8 \\ 8 & 1 & 4 \end{bmatrix}$$

ekspansi kofaktor baris pertama

$$\det(A) = 2\begin{vmatrix} 3 & 8 \\ 1 & 4 \end{vmatrix} - 8\begin{vmatrix} 1 & 8 \\ 8 & 4 \end{vmatrix} - 2\begin{vmatrix} 1 & 3 \\ 8 & 1 \end{vmatrix}$$

C_{11} : $(3 \times 4 - 8 \times 1)$
$= 12 - 8 = 4$

C_{12} : $(1 \times 4 - 8 \times 8)$
$= 4 - 64 = -60$

C_{13} : $(1 \times 1 - 3 \times 8)$
$= 1 - 24 = -23$

$\det(A) = 2(4) - 8(-60) + 2(-23)$
$= 8 + 480 - 46$
$= 442$ //

b) • $M_{11} \rightarrow \begin{vmatrix} 3 & 8 \\ 1 & 4 \end{vmatrix}$
$= (3 \times 4 - 8 \times 1)$
$= 12 - 8 = 4$

• $M_{21} \rightarrow \begin{vmatrix} 8 & 2 \\ 1 & 4 \end{vmatrix}$
$= (8 \times 4 - 2 \times 1)$
$= 32 - 2 = 30$

• $M_{31} \rightarrow \begin{vmatrix} 8 & 2 \\ 3 & 8 \end{vmatrix}$
$= (8 \times 8 - 2 \times 3)$
$= 64 - 6 = 58$

• $M_{12} \rightarrow \begin{vmatrix} 1 & 8 \\ 8 & 4 \end{vmatrix}$
$= (1 \times 4) - (8 \times 8)$
$= 4 - 64 = 60$

• $M_{22} \rightarrow \begin{vmatrix} 2 & 2 \\ 8 & 4 \end{vmatrix}$
$= (2 \times 4 - 8 \times 2)$
$= 8 - 16 = -8$

• $M_{32} \rightarrow \begin{vmatrix} 2 & 2 \\ 1 & 8 \end{vmatrix}$
$= (2 \times 8 - 2 \times 1)$
$= 16 - 2 = 14$

• $M_{13} \rightarrow \begin{vmatrix} 1 & 3 \\ 8 & 1 \end{vmatrix}$
$= (1 \times 1 - 3 \times 8)$
$= 1 - 24 = -23$

• $M_{23} \rightarrow \begin{vmatrix} 2 & 8 \\ 8 & 1 \end{vmatrix}$
$= (2 \times 1 - 8 \times 8)$
$= 2 - 64 = -62$

• $M_{33} \rightarrow \begin{vmatrix} 2 & 8 \\ 1 & 3 \end{vmatrix}$
$(2 \times 3 - 8 \times 1)$
$= 6 - 8 = -2$

Tabel ->

	j = 1	j = 2	j = 3
i = 1	4	-60	-23
i = 2	30	-8	-62
i = 3	58	14	-2

$$M = \begin{bmatrix} 4 & -60 & -23 \\ 30 & -8 & -62 \\ 58 & 14 & -2 \end{bmatrix}$$

c)

$$M = \begin{bmatrix} 4 & -60 & -23 \\ 30 & -8 & -62 \\ 58 & 14 & -2 \end{bmatrix}$$

$\det(A) = 442$

- matriks kofaktor

$$\begin{bmatrix} + & - & + \\ - & + & - \\ + & - & + \end{bmatrix} \rightarrow C = \begin{bmatrix} +4 & -(-60) & +(-23) \\ -(30) & +(-8 & -(-62) \\ +58 & -(14) & +(-2) \end{bmatrix}$$

$$= \begin{bmatrix} 4 & 60 & -23 \\ -30 & -8 & +62 \\ 58 & -14 & -2 \end{bmatrix}$$

- matriks adjoin

$$Adj(A) = \begin{bmatrix} 4 & -30 & 58 \\ 60 & -8 & -14 \\ -23 & 62 & -2 \end{bmatrix}$$

- inverse matrikes

$$A^{-1} = \frac{1}{442} \cdot Adj(A)$$

$$= \frac{1}{442} \cdot \begin{bmatrix} 4 & -30 & 58 \\ 60 & -8 & -14 \\ -23 & 62 & -2 \end{bmatrix}$$

$$A^{-1} = \begin{bmatrix} \frac{4}{442} & \frac{-30}{442} & \frac{58}{442} \\ \frac{60}{442} & \frac{-8}{442} & \frac{-14}{442} \\ \frac{-23}{442} & \frac{62}{442} & \frac{-2}{442} \end{bmatrix}$$

d) MATLAB

② e)

$$P = \begin{vmatrix} 120 & (30+N) & 200 \\ 80 & 0+2 & (10+N) \\ (30+N) & 90 & 170 \end{vmatrix}$$

$$P = \begin{vmatrix} 120 & 30+18 & 200 \\ 80 & 1+2 & 10+18 \\ 30+18 & 90 & 170 \end{vmatrix}$$

$$P = \begin{bmatrix} 120 & 68 & 200 \\ 80 & 3 & 28 \\ 48 & 90 & 170 \end{bmatrix} \qquad A = \begin{bmatrix} 2 & 8 & 2 \\ 1 & 3 & 8 \\ 8 & 1 & 4 \end{bmatrix}$$

- enkripsi: $E = A.P$

- $E_{11} = (2\times120) + (8\times80) + (2\times48)$
  $= 240 + 640 + 96 = 976$

- $E_{12} = (2\times68) + (8\times3) + (2\times90)$
  $= 136 + 24 + 180 = 340$

- $E_{13} = (2\times200) + (8\times28) + (2\times170)$
  $= 400 + 224 + 340 = 964$

- $E_{21} = (1\times120) + (3\times80) + (8\times48)$
  $= 120 + 240 + 384 = 744$

- $E_{22} = (1\times68) + (3\times3) + (8\times90)$
  $= 68 + 9 + 720 = 797$

- $E_{23} = (1\times200) + (3\times28) + (8\times170)$
  $= 200 + 84 + 1360 = 1644$

- $E_{31} = (8\times120) + (1\times80) + (4\times48)$
  $= 960 + 80 + 192 = 1232$

- $E_{32} = (8\times68) + (1\times3) + (4\times90)$
  $= 544 + 3 + 360 = 907$

- $E_{33} = (8\times200) + (1\times28) + (4\times170)$
  $= 1600 + 28 + 680 = 2308$

$$E = \begin{bmatrix} 976 & 340 & 964 \\ 744 & 797 & 1644 \\ 1232 & 907 & 2308 \end{bmatrix}$$

· deskripsi $P' = A' \cdot E$

$$A^{-1} = \frac{1}{442}\begin{bmatrix} 4 & -30 & 58 \\ 60 & -8 & -14 \\ -23 & 62 & -2 \end{bmatrix}$$

· $P'_{11} \Rightarrow \frac{1}{442}(4 \times 976 - 30 \times 744 + 58 \times 1232)$

$= \frac{1}{442}(3904 - 22320 + 71456) = \frac{53040}{442} = 120$

· $P'_{21} = \frac{1}{442}(60 \times 976 - 8 \times 744 - 14 \times 1232)$

$= \frac{1}{442}(58560 - 5952 - 17248) = \frac{35360}{442} = 80$

· $P'_{22} = \frac{1}{442}(60 \times 340 - 8 \times 797 - 14 \times 907)$

$= \frac{1}{442}(20400 - 6376 - 12698) = \frac{1326}{442} = 3$

· $P'_{12} = \frac{1}{442}(4 \times 340 - 30 \times 797 + 58 \times 907)$

$= \frac{1}{442}(1360 - 23910 + 52606) = \frac{30056}{442} = 68$

· $P'_{13} = \frac{1}{442}(4 \times 964 - 30 \times 1644 + 58 \times 2308)$

$= \frac{1}{442}(3856 - 49320 + 133864) = \frac{88400}{442} = 200$

· $P'_{23} = \frac{1}{442}(60 \times 964 - 8 \times 1644 - 14 \times 2308)$

$= \frac{1}{442}(57840 - 13152 - 32312) = \frac{12376}{442} = 28$

· $P'_{31} = \frac{1}{442}(-23 \times 976 + 62 \times 744 - 2 \times 1232)$

$= \frac{1}{442}(-22448 + 46128 - 2464) = \frac{21216}{442} = 48$

· $P'_{32} = \frac{1}{442}(-23 \times 340 + 62 \times 797 - 2 \times 907)$

$= \frac{1}{442}(-7820 + 49414 - 1814) = \frac{39780}{442} = 90$

② $P'_{33} = \frac{1}{442}(-28 \times 964 + 62 \times 1644 - 2 \times 2308)$

$= \frac{1}{442}(-22172 + 101928 - 4616) = \frac{75140}{442} = 170$

$$P' = \begin{bmatrix} 120 & 68 & 200 \\ 80 & 3 & 28 \\ 48 & 90 & 170 \end{bmatrix} \rightarrow P' = P$$

Diskusi : Apa yang terjadi jika det (A) = 0 ?

Jawab : Jika det(A) = 0 , maka matriks A tidak memiliki invers sehingga proses dekripsi tidak dapat dilakukan . Akibatnya, data citra yang telah di enkripsi tidak dapat dikembalikan ke bentuk semula ($P' \neq P$). Hal ini menyebabkan kehilangan informasi dan sistem enkripsi menjadi tidak valid